



## Guidelines

# Sodium Glucose Co-Transporter2 (SGLT2) - Inhibitors (Gliflozins) And Euglycaemic Ketoacidosis

## Scope

| Site     | Service/Department/Unit | Disciplines                |
|----------|-------------------------|----------------------------|
| SCGOPHCG | Organisation Wide       | Medical, Nursing, Pharmacy |

## Introduction

This guideline provides guidance to all SCGHOPHCG staff involved in the prescription, administration, supply and management of patients on Sodium-Glucose co-Transporter-2 inhibitors (SGLT2 inhibitors) and euglycaemic ketoacidosis. This document supports and complies with related legislation, policy and other related documents [here](#).

### 1.0 General Information<sup>1,3</sup>

SGLT2 inhibitors (gliflozins) are increasingly prescribed to manage diabetes, heart failure and kidney disease.

Ketoacidosis has been associated with the use of SGLT2 Inhibitors in diabetes and there is a small but definitive risk of severe diabetic ketoacidosis (DKA) with near normal or only mildly elevated blood glucose levels (ie 'euglycaemic' ketoacidosis); therefore, a normal or only modestly elevated plasma glucose level does not exclude the diagnosis.

The risk is increased if:

- the patient is fasting or has a restricted dietary (especially carbohydrate) intake,
- has undergone bowel preparation and/or a surgical procedure,
- is dehydrated,
- has an active infection,
- has a history of active drug or alcohol abuse
- a missed or reduced insulin dose.

These should be considered for patients taking SGLT2 inhibitors in management of heart failure and kidney disease as well.

In the management of diabetes, gliflozins are ordinarily prescribed to reduce chronic complications of diabetes rather than to manage acute hyperglycaemia, and withholding the drug is not likely to cause significant glucose instability. Short-term insulin therapy should be used as the most appropriate means of maintaining stable glucose levels peri-operatively or during hospitalisation for an acute illness if required.

Routine blood glucose and blood ketone testing is recommended in diabetic patients admitted to hospital who have been on SGLT2 inhibitors and who are unwell, fasting (including in the peri-operative state) or with limited oral intake. Ketones should also be measured in those treated with SGLT2 inhibitors who are, or are suspected of being, insulin deficient (eg: Type I diabetes, HbA1c >9%, pancreatic pathology).

## 2.0 Presentation

- Empagliflozin and dapagliflozin are the only gliflozins available in Australia. They are available in a variety of combination tablets, listed in the table below.
- Refer to [SCGOPHCG approved references](#) (e.g. Australian Medicines Handbook) for more information.
- There are other gliflozins available outside of Australia.

## 3.0 Features of DKA/euglycaemic DKA (euDKA)

- DKA/euDKA should be considered in patients with diabetes taking SGLT2 inhibitors who have any of the following features:
  - Drowsiness, abdominal pain, confusion, extreme thirst, nausea, vomiting, fatigue or unexplained metabolic acidosis
  - Finger prick capillary blood ketone (or blood beta-hydroxybutyrate) levels >0.6mmol/L in the perioperative period or >1.5mmol/L at any other time.
  - Metabolic acidosis on VBG or ABG (typically arterial pH ≤7.3, serum bicarbonate ≤ 15mmol/L and anion gap >12mmol/L)
- **Note:**
  - a normal plasma glucose level does not exclude the diagnosis
  - normal urine ketones do not exclude the diagnosis
  - lactic acidosis is an important differential diagnosis, but this condition may also precipitate euglycaemic ketoacidosis

## 4.0 Management

The table below contains guidance on when to withhold gliflozins and when to consider blood ketone testing for various patient groups.

| Patient Group | Guidance for Management                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|               | <ul style="list-style-type: none"> <li>• Gliflozin should be <b>withheld at least 3 days pre-operatively</b> (2 days prior to surgery and the day of surgery.)</li> <li>• Strongly consider postponing non-urgent surgery if gliflozins have not been ceased 3 to 4 days prior to surgery and either blood ketones are &gt;0.6mmol/L or HbA1c is &gt;9.0%, as these are indicators of insulin insufficiency and a higher risk of DKA</li> <li>• Monitor <b>BLOOD</b> glucose and <b>BLOOD</b> ketones as clinically appropriate, i.e.:               <ul style="list-style-type: none"> <li>○ on admission</li> <li>○ in the setting of significant hyperglycaemia (BGL &gt; 15mmol/L)</li> <li>○ if clinically unwell</li> </ul> </li> </ul> |

|                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
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| Non-urgent surgery                                                                                                                                                | <ul style="list-style-type: none"> <li>• <b>Do not restart gliflozin until patient has recovered</b> from the operation, is well hydrated and has resumed eating normally. This will likely not be until several days after surgery.</li> <li>• An alternative therapy may need to be prescribed for the period of withholding if hyperglycaemia is present.</li> </ul>                                                                                                                                                                                                                                                                                                                                         |
| Urgent Surgery                                                                                                                                                    | <ul style="list-style-type: none"> <li>• <b>Withhold gliflozin IMMEDIATELY.</b></li> <li>• Monitor <b>BLOOD</b> glucose and <b>BLOOD</b> ketones as clinically appropriate, i.e.:               <ul style="list-style-type: none"> <li>○ on admission</li> <li>○ in the setting of significant hyperglycaemia (BGL &gt; 15mmol/L)</li> <li>○ if clinically unwell</li> </ul> </li> <li>• <b>Do not restart gliflozin until patient has physiologically improved</b>, is well hydrated and resumed eating normally. This will likely not be until several days after surgery.</li> <li>• An alternative therapy may need to be prescribed for the period of withholding if hyperglycaemia is present.</li> </ul> |
| Other patients: <ul style="list-style-type: none"> <li>- with restricted dietary intake</li> <li>- who are dehydrated</li> <li>- with active infection</li> </ul> | <ul style="list-style-type: none"> <li>• <b>Consider withholding gliflozin</b></li> <li>• Monitor <b>BLOOD</b> glucose and <b>BLOOD</b> ketones as clinically appropriate, i.e.:               <ul style="list-style-type: none"> <li>○ on admission</li> <li>○ in the setting of significant hyperglycaemia (BGL &gt; 15mmol/L)</li> <li>○ if clinically unwell</li> </ul> </li> <li>• <b>Do not restart gliflozin until patient has physiologically improved</b>, is well hydrated and resumed eating normally.</li> <li>• In diabetic patients, an alternative therapy may need to be prescribed for the period of withholding if hyperglycaemia is present.</li> </ul>                                      |

## 5.0 Practical points

- Contact the treating medical officer for assistance if blood (capillary) ketone levels are above **0.6 mmol/L**.
- Include management instructions to patients and GP in the discharge summary e.g.: instructions to restart gliflozins when recovered and back on normal dietary intake.

- Educate patients about the reason why gliflozins need to be withheld when oral intake is significantly decreased.

## Related Documents

### Legislation:

- [Medicines and Poisons Act 2014](#)
- [Medicines and Poisons Regulations 2016](#)
- [Health Services Act 2016](#)

### Authorising policy and standards

- [SCGOPHCG Medicines Management Policy and Procedure](#)
- [SCGOPHCG High Risk Medication Policy](#)
- [Mandatory Policy MP0077/18 Statewide Medicines Formulary Policy](#)
- [OD 0647/16 National Standard for User-Applied Labelling of Injectable Medicines, Fluids and Lines 2016](#)
- [Australian Commission on Safety and Quality in Healthcare. Recommendation for terminology, abbreviations and symbols used in medicines documentation, December 2016](#)
- NSQHS Standards: 1,2,4,6,8

### Supporting documents

- Adult Diabetic Ketoacidosis (DKA) Guidelines and Management Record (MR 849) SCGH
- Adult Variable Rate Insulin Infusion Guideline and Management Record (MR826) SCGH
- Adult Hyperosmolar Hyperglycaemic State Guideline and Management Record (MR 848) SCGH

## References

1. Australian Medicines Handbook [Internet]. Adelaide (SA): Australian Medicines Handbook Pty Ltd; Sodium-glucose co-transporter 2 inhibitor [cited 2024 May 28]. Available from: Australian Medicines Handbook.
2. Australian Diabetes Society Alert – Severe Euglycaemic Ketoacidosis with SGLT2 Inhibitor Use in the Perioperative Period
3. ANZCA Safety Alert [https://www.anzca.edu.au/getattachment/57adf09c-68c2-4469-8782-2978453a58df/Periprocedural-Diabetic-Ketoacidosis-\(DKA\)-with-SGLT2-Inhibitor-Use](https://www.anzca.edu.au/getattachment/57adf09c-68c2-4469-8782-2978453a58df/Periprocedural-Diabetic-Ketoacidosis-(DKA)-with-SGLT2-Inhibitor-Use)
4. Euglycaemic diabetic ketoacidosis associated with sodium-glucose cotransporter-2 inhibitors: New drugs bring new problems Australasian Anaesthesia 2019 pp252 – 265 <http://www.anzca.edu.au/documents/blue-book-2019.pdf>
5. Venkatesan Thiruvankatarajan, Emily Jane Meyer, Nagesh Nanjappa, Roelof M. Van Wijk and David Jesudason. Perioperative diabetic ketoacidosis associated with sodium-glucose co-transporter-2 inhibitors: a systematic review. British Journal of Anaesthesia, 123 (1): 27e36 (2019)
6. AusDI by Telstra Health [Internet]. Empagliflozin (Systemic) Independent Drug Monograph [cited 2024 May 28]. Available from: AusDI.
7. DeSantis A, Nathan DM, Rubinow K (2024). Sodium-glucose cotransporter 2 inhibitors for the treatment of hyperglycemia in type 2 diabetes mellitus. Up-to-date Inc. [cited 2024 May 28]. Available from: [Sodium-glucose cotransporter 2 inhibitors for the treatment of hyperglycemia in type 2 diabetes mellitus - UpToDate \(health.wa.gov.au\)](#)
8. Muluk V et al. (2024). Perioperative medication management. Up-to-date Inc. [cited 2024 May 28]. Available from: [Perioperative medication management - UpToDate \(health.wa.gov.au\)](#)

## Authorisation & Version Control

|                      |                                                                                                                                                                                                                                                              |                |            |              |            |
|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|------------|--------------|------------|
| Policy Sponsor       | SCGOPHCG Director of Clinical Services                                                                                                                                                                                                                       |                |            |              |            |
| Policy Contact       | SCGOPHCG Medicines and Therapeutics Committee Executive Officer (Senior Pharmacist)                                                                                                                                                                          |                |            |              |            |
| Date First Issued:   | 11/03/2019                                                                                                                                                                                                                                                   | Last Reviewed: | 09/01/2025 | Review Date: | 09/01/2029 |
| Version No.          | 4                                                                                                                                                                                                                                                            |                |            |              |            |
| Approved by:         | SCGOPHCG Medicines and Therapeutics Committee                                                                                                                                                                                                                |                |            | Date:        | 09/01/2029 |
| Acknowledgments:     | V1 acknowledgements: SCGH Pharmacy Department, Head of Department Diabetes Service Endocrinology, Head of Department Anaesthetics, Chair Nursing Practice Guidelines, CNC Diabetes<br>V2 acknowledgements: SCGH Pharmacy Department, OPH Pharmacy Department |                |            |              |            |
| Standards Applicable | NSQHS Standards (Version 2)                                                                                                                                                |                |            |              |            |

For full details regarding, development, review, consultation, approval, endorsement - contact the Policy Officer for the submission form

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